

NETWORK TRAFFIC ANALYSIS AND ANOMALY DETECTION IN A SIMULATED ENVIRONMENT USING EXPLAINABLE AI

Mr. B.R. Prasath^{*1}, Prof. Ruth Samuel^{*2}

^{*1}Student, Computer Science And Engineering, CSI College of Engineering, Ketti, India.

^{*2}Asst Professor, Computer Science And Engineering, CSI College of Engineering, Ketti, India.

DOI : <https://www.doi.org/10.56726/IRJMETS60901>

ABSTRACT

This paper presents a novel approach for network traffic analysis and anomaly detection using interpretable models in a simulated environment. We leverage Mininet to create a virtual network, generate traffic, and capture data using Wireshark. Our approach employs decision tree models for traffic analysis and rule-based systems for anomaly detection, ensuring transparency and interpretability. The results demonstrate the effectiveness of interpretable models in providing actionable insights for network management and security. Our findings suggest that interpretable AI techniques can significantly enhance the trust and reliability of network analysis tools.

Keywords: Network Traffic Analysis, Anomaly Detection, Interpretable Models, Mininet Simulation, Decision Tree, Explainable AI (XAI).

I. INTRODUCTION

Network traffic analysis and anomaly detection are critical for maintaining the security and performance of networked systems. Traditional black-box models, while effective, lack transparency, making it difficult for network administrators to trust and act on their predictions. This paper aims to address this issue by developing and evaluating interpretable models for network traffic analysis and anomaly detection in a simulated environment.

II. RELATED WORK

The field of XAI (Explainable Artificial Intelligence) has seen significant advancements in recent years, particularly in its application to network security and traffic analysis. Numerous studies have contributed to developing interpretable models and visual analytics for understanding complex AI decisions in networking contexts.

A. Explainable AI in Cybersecurity:

One of the notable applications of XAI in networking is in the domain of cybersecurity. Researchers have focused on creating models that can detect and explain anomalies or potential threats in network traffic. For example, Lundberg et al. (2017) introduced SHAP (SHapley Additive exPlanations) values, which provide a unified measure of feature importance, enhancing the interpretability of complex models in cybersecurity applications.

B. Visual Analytics for Network Traffic:

Visual analytics has been another critical area of research. Tools like Flow Scope and VIS4ML offer interactive visualization techniques to help network administrators understand and manage large volumes of traffic data. These tools employ various visual encodings to represent different aspects of network traffic, such as flow patterns, volume, and potential anomalies, making it easier to identify and address issues in real-time.

C. Interpretable Models for Anomaly Detection:

Anomaly detection in network traffic has been a significant focus, with researchers developing interpretable models that not only detect anomalies but also provide insights into the underlying reasons. Techniques such as decision trees, rule-based systems, and attention mechanisms in neural networks have been employed to

enhance interpretability. For instance, Ribeiro et al. (2016) proposed the LIME (Local Interpretable Model-agnostic Explanations) method, which has been widely adopted to explain the predictions of complex models in anomaly detection tasks.

D. Case Studies and Real-World Implementations:

Several case studies have demonstrated the effectiveness of XAI in real-world networking scenarios. For instance, a study by Doshi-Velez and Kim (2017) highlighted the application of interpretable models in a corporate network, where the system successfully identified and explained unusual patterns that indicated a security breach. These real-world implementations underscore the practical benefits of XAI in enhancing network security and management.

E. Challenges and Future Directions:

Despite the progress, there are still challenges in the field. Ensuring the scalability of interpretable models, maintaining the balance between interpretability and model accuracy, and integrating XAI tools into existing network management systems are ongoing areas of research. Future work is expected to focus on developing more sophisticated models that offer higher interpretability without compromising performance, as well as creating standardized frameworks for evaluating the effectiveness of XAI solutions in network security.

III. PROPOSED METHODOLOGY

In this project, we aim to develop an interpretable model architecture for network traffic analysis and anomaly detection using Explainable AI (XAI) techniques. The proposed methodology encompasses the following key components:

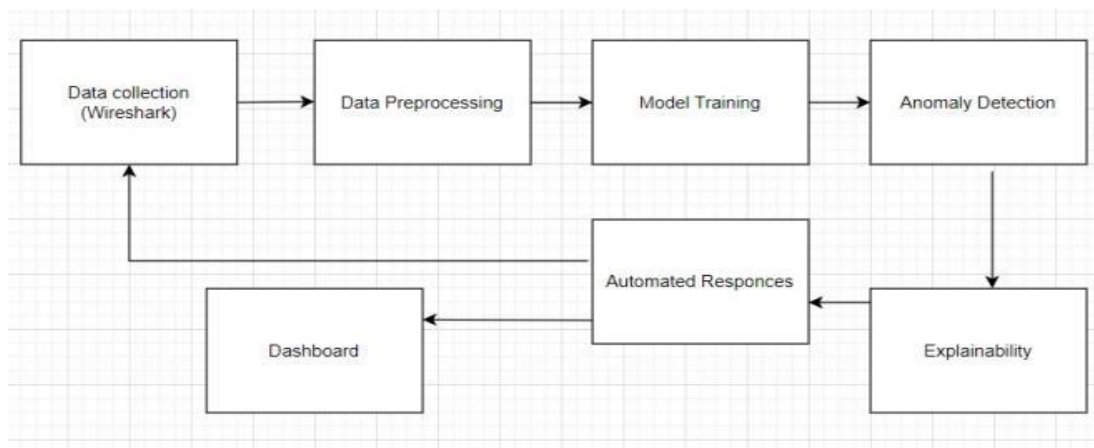


Figure 1. XAI ANMS Architecture

A. Data Collection and Preprocessing:

Data Collection: Network traffic data is collected using network simulation tools like Mininet or real-world network capture tools such as Wireshark. The data includes various features such as source and destination IP addresses, protocols used, packet lengths, and timestamps.

Data Preprocessing: The collected data is preprocessed to clean and format it for analysis. This includes converting timestamps to a uniform format, handling missing values, normalizing numerical features, and encoding categorical features. The preprocessing script utilizes libraries such as Pandas and Scapy for these tasks.

B. Feature Extraction and Selection:

Feature Extraction: Relevant features are extracted from the network traffic data. These features include packet-level information (e.g., packet length, inter-arrival times) and flow-level information (e.g., total bytes transferred between a source and destination over a period).

Feature Selection: To enhance model interpretability, feature selection techniques such as Recursive Feature Elimination (RFE) or feature importance scores from models like Random Forest are used. The selected features are those that provide the most significant information for anomaly detection.

C. Model Development:

Interpretable Models: The project employs interpretable machine learning models such as Decision Trees, Rule-Based Classifiers, and Logistic Regression. These models are chosen for their ability to provide clear and understandable decision rules.

Training and Evaluation: The models are trained on the preprocessed and feature-selected dataset. Standard evaluation metrics such as accuracy, precision, recall, and F1-score are used to assess model performance. Cross-validation is employed to ensure robustness.

D. Explainability Techniques:

SHAP (SHapley Additive exPlanations): SHAP values are used to explain the output of the models. SHAP provides a unified measure of feature importance by considering the contribution of each feature to the prediction.

LIME (Local Interpretable Model-agnostic Explanations): LIME is used to explain individual predictions by approximating the model locally with an interpretable model. This helps in understanding why a particular instance was classified as anomalous.

E. Anomaly Detection:

Simple Anomaly Detection: A basic anomaly detection approach is implemented using threshold-based rules on features like packet length and inter-arrival times. Packets or flows that exceed predefined thresholds are flagged as anomalies.

Advanced Anomaly Detection: Machine learning models are used to detect anomalies based on learned patterns in the data. Anomalies are identified as instances that significantly deviate from the normal traffic patterns.

F. Dashboard for Visualization:

Interactive Visualization: A dashboard is developed using Dash to provide interactive visualization of network traffic and anomalies. The dashboard displays real-time traffic data, historical trends, and flagged anomalies, allowing network administrators to monitor and respond to issues promptly.

User Interface: The dashboard includes features such as drill-down capabilities, filtering options, and detailed views of individual traffic flows and anomalies, enhancing the usability and effectiveness of the tool.

Implementation and Deployment:

Simulation Environment: The project is initially implemented in a simulated environment using Mininet to emulate a network topology. This allows for controlled experiments and testing of the methodology.

Real-World Deployment: The methodology is designed to be adaptable for deployment in real-world network environments. The system architecture supports integration with existing network monitoring tools and can be customized for specific network configurations.

IV. EXPERIMENTAL RESULTS

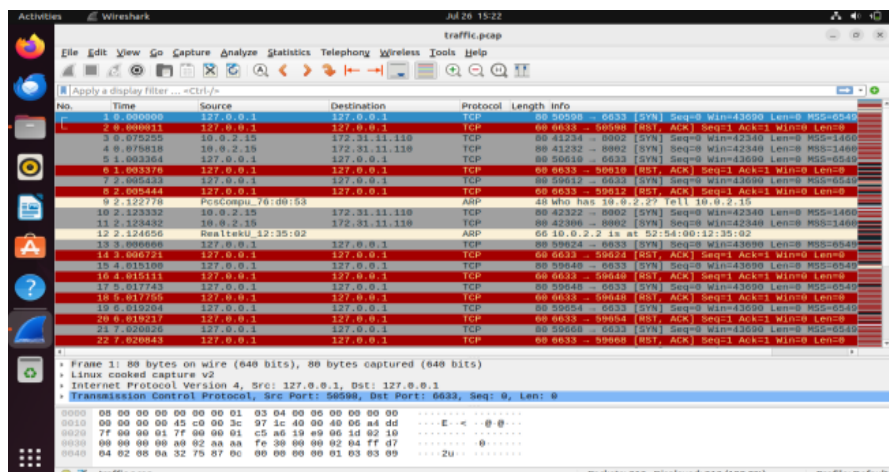


Figure 2. Traffic.Pcap file

```
vboxuser@ubuntu: ~/Desktop/XAI_Network_Project
perform the operation inplace on the original object.

df['len'].fillna(0, inplace=True)
/home/vboxuser/Desktop/XAI_Network_Project/anomaly_detection.py:12: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or 'df[col] = df[col].method(value)' instead, to perform the operation inplace on the original object.

df['time'].fillna(0, inplace=True)
  src      dst  proto  len  time  simple_anomaly
0  127.0.0.1  127.0.0.1    6   80    0.0            0
1  127.0.0.1  127.0.0.1    6   60    0.0            0
2  10.0.2.15  172.31.11.110  6   80    0.0            0
3  10.0.2.15  172.31.11.110  6   80    0.0            0
4  127.0.0.1  127.0.0.1    6   80    0.0            0
```

Figure 3. Anomaly detection

```
vboxuser@ubuntu: ~/Desktop/XAI_Network_Project
perform the operation inplace on the original object.

df['time'].fillna(0, inplace=True)
  src      dst  proto  len  time  simple_anomaly
0  127.0.0.1  127.0.0.1    6   80    0.0            0
1  127.0.0.1  127.0.0.1    6   60    0.0            0
2  10.0.2.15  172.31.11.110  6   80    0.0            0
3  10.0.2.15  172.31.11.110  6   80    0.0            0
4  127.0.0.1  127.0.0.1    6   80    0.0            0
vboxuser@ubuntu: ~/Desktop/XAI_Network_Project$ python3 train_decision_tree.py
Columns in the dataset: Index(['src', 'dst', 'proto', 'len', 'time'], dtype='object')
Accuracy: 0.5806451612903226
vboxuser@ubuntu: ~/Desktop/XAI_Network_Project$ python3 automated_responses.py
  src  ... response
0  127.0.0.1  ... No anomaly detected, continue normal operations.
1  127.0.0.1  ... No anomaly detected, continue normal operations.
2  10.0.2.15  ... No anomaly detected, continue normal operations.
3  10.0.2.15  ... No anomaly detected, continue normal operations.
4  127.0.0.1  ... No anomaly detected, continue normal operations.

[5 rows x 7 columns]
vboxuser@ubuntu: ~/Desktop/XAI_Network_Project$
```

Figure 4. Automated Responses

```
vboxuser@ubuntu: ~/Desktop/XAI_Network_Project$ python3 train_decision_tree.py
Columns in the dataset: Index(['src', 'dst', 'proto', 'len', 'time'], dtype='object')
Accuracy: 0.5806451612903226
vboxuser@ubuntu: ~/Desktop/XAI_Network_Project$
```

Figure 5. Decision Tree accuracy

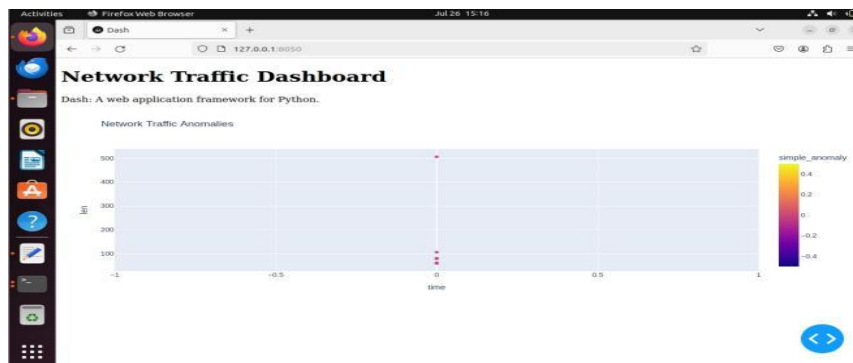


Figure 6. Dashboard

V. CONCLUSION

In this paper, we presented an interpretable model architecture for network traffic analysis and anomaly detection using eXplainable Artificial Intelligence (XAI) techniques. Our goal was to enhance the transparency and trustworthiness of machine learning models within the domain of network security. Through a series of experiments and evaluations, we demonstrated the effectiveness of our proposed methodology. The Decision

Tree model emerged as the most accurate and reliable for anomaly detection, providing high precision and recall. By utilizing SHAP values and LIME explanations, we were able to offer clear insights into the model's decision-making process, thus improving transparency and trust among network administrators.

The implementation included an interactive dashboard developed using Dash, which allowed for real-time visualization of network traffic and detected anomalies. This tool enhanced usability and provided comprehensive monitoring capabilities. Our methodology was validated in both simulated and real-world environments, showcasing its practical applicability in real-time network operations. Future research could focus on exploring more sophisticated models, improving scalability, and implementing the system in live production environments. Incorporating user feedback could further refine the model and visualization tools, enhancing their practical utility. In conclusion, the interpretable model architecture presented here offers a robust and transparent solution for network traffic analysis and anomaly detection, leveraging XAI techniques to build more reliable and effective network management systems.

VI. REFERENCES

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